

DeFacto: Counterfactual Thinking with Images for Enforcing Evidence-Grounded and Faithful Reasoning

Tianrun Xu^{1,2} Haoda Jing¹ Ye Li³ Yuquan Wei⁴ Jun Feng^{2,5} Guanyu Chen¹ Haichuan Gao⁶
Tianren Zhang¹ Jing Liu^{2,5,7} Feng Chen^{†,1}

Abstract

Recent advances in multimodal language models (MLLMs) have made *thinking with images* a dominant paradigm for multimodal reasoning. However, existing methods still fail to ensure *evidence-answer consistency*, where correct answers must be supported by correct visual evidence. To address this issue, we propose *DeFacto*, a counterfactual reasoning framework that explicitly aligns visual evidence with final answers. Our approach integrates three complementary training paradigms: (i) positive, (ii) counterfactual, and (iii) random-masking. We further develop a language-guided evidence construction pipeline that automatically localizes question-relevant regions and generates counterfactual variants, resulting in **DeFacto-100K**. Building on this dataset, we train MLLMs with GRPO-based reinforcement learning and design three complementary rewards to promote correct answering, structured reasoning, and consistent evidence selection. Moreover, we introduce **DeFacto-1.5K**, a human-annotated benchmark for systematically evaluating evidence-grounded consistency beyond answer accuracy. Experiments on diverse benchmarks demonstrate that *DeFacto* substantially improves both answer accuracy and evidence-answer consistency over strong baselines. The code and datasets are available at <https://github.com/tinnel123666888/defacto>.

1. Introduction

Vision-language models (VLMs) (Alayrac et al., 2022; Li et al., 2023a; Zhu et al., 2023; Liu et al., 2023; 2024a; Peng et al., 2023; Bai et al., 2023; Team et al., 2023; Chen et al., 2024c; Xu et al., 2025a; Yan et al., 2026) have achieved remarkable progress in recent years, demonstrating strong capabilities across a wide range of multimodal tasks such as visual question answering, image captioning, and referring expression comprehension. By leveraging large-scale pretraining and cross-modal alignment, these models can generate fluent and semantically relevant outputs grounded in visual context. However, in complex scenarios that require multi-step reasoning or fine-grained perception, existing models often rely heavily on implicit language priors, producing plausible yet unfaithful responses that are weakly grounded in the actual image. Instead of genuinely learning to reason over visual content, these models often fall back on text-based chain-of-thought patterns, limiting their ability to handle cases where critical evidence must be directly perceived from the image.

Recent advances in “thinking with images” (Microsoft, 2024; OpenAI, 2025) integrate explicit visual steps into multimodal reasoning to improve interpretability and visual grounding. Early approaches adopt supervised fine-tuning (SFT) (Ouyang et al., 2022; Touvron et al., 2023; Liu et al., 2023; Dettmers et al., 2023; Xu et al., 2025b), where models learn to generate region-aware reasoning traces in a chain-of-thought manner with manually annotated visual steps (Shao et al., 2024). To reduce annotation cost, subsequent works explore reinforcement learning to enable autonomous visual interactions such as cropping, attention shifting, and zooming (Zheng et al., 2025; Cao et al., 2025; Liu et al., 2025b; Zhang et al., 2025b; Su et al., 2025). Yet existing thinking-with-images methods still cannot guarantee *evidence-answer consistency*, often exhibiting three failure cases: (i) **Mislocalized Failure** (wrong evidence, wrong answer), (ii) **Spurious Correctness** (wrong evidence, correct answer), and (iii) **Faithful Incorrectness** (correct evidence, wrong answer). As shown in Fig. 1, we use two representative baselines, DeepEyes and GRIT, to qualitatively illustrate these issues. These observations suggest that reliable mul-

¹Department of Automation, Tsinghua University, Beijing, China ²Zhongguancun Academy, Beijing, China ³School of Software, Xinjiang University, Urumqi, China ⁴College of Materials Science and Engineering, Fuzhou University, Fuzhou, China ⁵Institute of Automation, Chinese Academy of Sciences, Beijing, China ⁶Beijing Qianjue Technology Co., Ltd., Beijing, China ⁷School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing, China. Correspondence to: Feng Chen <chenfeng@mail.tsinghua.edu.cn>.

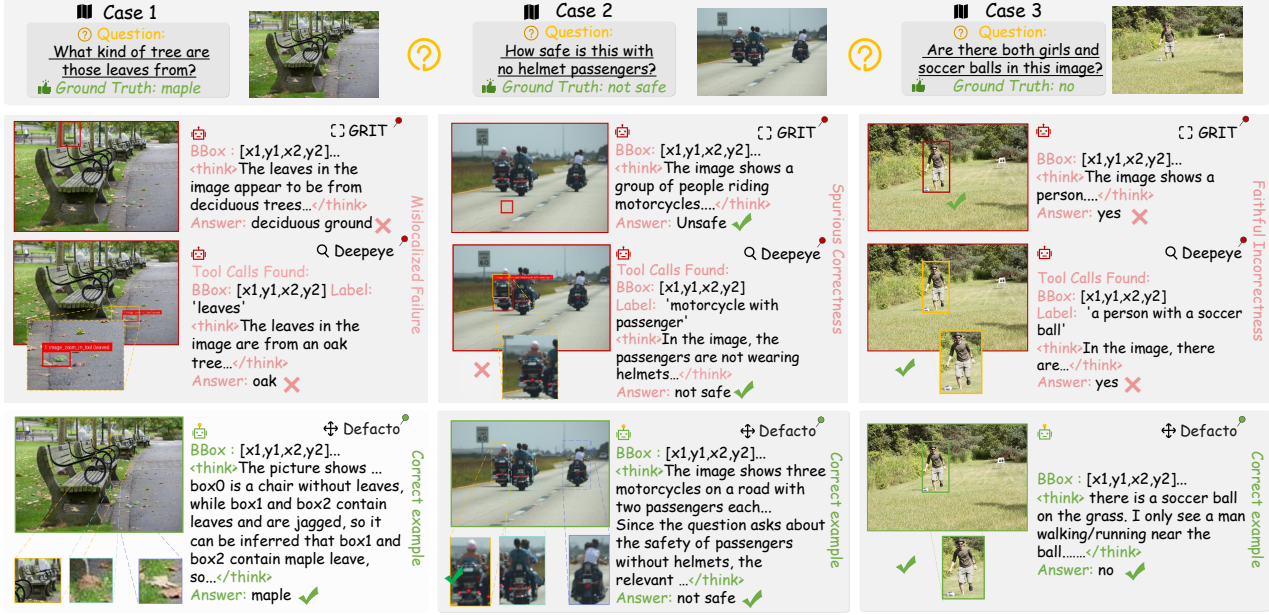


Figure 1. Qualitative examples of **unfaithful reasoning** in thinking-with-images. We categorize failures into three types: *Mislocalized Failure* (wrong evidence, wrong answer), *Spurious Correctness* (wrong evidence, correct answer), and *Faithful Incorrectness* (correct evidence, wrong answer). We compare two representative baselines (Deepeyes and GRIT) with our method on three cases.

timodal reasoning requires jointly enforcing both correct answering and evidence-aligned visual reasoning, rather than optimizing accuracy alone.

Motivated by these failure cases, we introduce *DeFacto*, a counterfactual reasoning framework that aligns reasoning trajectories with visual evidence to improve evidence-answer consistency. The core idea is to jointly constrain the model via three complementary training forms: (i) **positive supervision**, (ii) **counterfactual abstention**, and (iii) **random masking**. In the positive case, the model is trained on the original image to predict evidence regions $\mathcal{R}^+$ (bounding boxes covering essential visual cues) and the correct answer, receiving positive feedback only when both are correct. In the counterfactual case, the regions $\mathcal{R}^+$ are masked and the model is required to abstain by outputting a designated token (e.g., “unknown”). In the random-masking case, irrelevant regions $\mathcal{R}^-$ are masked to prevent shortcut learning and ensure the abstention behavior is triggered only by missing evidence rather than the presence of masks.

In practice, constructing counterfactual samples requires reliably identifying the question-relevant regions. To this end, we adopt a structured two-stage extraction pipeline. First, a multimodal language model (Qwen2.5-VL (Bai et al., 2025)) parses the question and generates a set of key descriptors (e.g., “the red cup,” “the text on the shirt”). Next, candidate regions in the image are obtained from a region proposal network (RPN) (Ren et al., 2015) and an OCR module (Islam et al., 2017). The OCR regions are further

matched with textual descriptors to capture evidence critical for text-centric questions. For visual objects, the descriptors are fed into an open-vocabulary detector (DINO-X (Ren et al., 2024)), which provides bounding boxes that serve as positive evidence regions. Finally, the remaining proposals from the RPN, after removing matched positives, are treated as irrelevant regions for counterfactual construction. Using this pipeline, we construct **DeFacto-100K**, a counterfactual dataset of about 100k images, ensuring that positive, counterfactual, and random-masking instances differ only in the availability of essential evidence while preserving unrelated context. Building on **DeFacto-100K**, the model is further optimized with GRPO-based reinforcement learning. This training paradigm enforces consistency between evidence selection and final predictions, ensuring that reasoning traces remain aligned with the visual evidence used to justify the answer. As illustrated in Fig. 1, our method selects regions that support the question and maintains coherent evidence-to-answer consistency.

Our main contributions are summarized as follows:

(1) We propose *DeFacto*, a counterfactual *thinking with images* framework that aligns the reasoning process with essential visual evidence by jointly optimizing answer correctness and region-level faithfulness via GRPO-based reinforcement learning.

(2) We construct **DeFacto-100K**, a counterfactual dataset of about 100k images built with a language-guided pipeline

that integrates open-vocabulary detection with targeted evidence masking, ensuring that only question-relevant regions are removed while irrelevant context is preserved.

(3) We introduce **DeFacto-1.5K**, a 1.5k-sample faithfulness benchmark with human-verified evidence annotations, enabling systematic evaluation of region-level reasoning faithfulness beyond answer accuracy.

(4) Through extensive experiments on diverse VQA benchmarks as well as **DeFacto-1.5K**, we demonstrate that *DeFacto* consistently improves both answer accuracy and grounding faithfulness over strong baselines.

Conflict of Interest Disclosure. The authors declare no financial conflicts of interest related to this work.

2. Related Work

Multimodal Large Language Models. Recent years have witnessed rapid progress in multimodal large language models (MLLMs), which extend large language models with visual encoders to support unified vision-language understanding and reasoning. Representative proprietary systems such as GPT-4V and GPT-4o (Microsoft, 2024) demonstrate strong multimodal capabilities and accelerate the trend of integrating explicit visual interaction into reasoning (OpenAI, 2025). On the open-source side, Qwen-VL/Qwen2.5-VL (Bai et al., 2025) provides scalable and strong multimodal backbones with competitive OCR and document understanding; LLaVA-style models (Liu et al., 2023) popularize instruction-tuned MLLMs by aligning vision encoders with LLMs through multimodal instruction data; and DeepSeek-VL (Wu et al., 2024) advances large-scale multimodal pretraining with competitive performance.

Counterfactual Reasoning in Vision-Language Models. Counterfactual reasoning in VLMs can be categorized into two types: counterfactual data generation and inference-based reasoning. The first enhances robustness by constructing or augmenting counterfactual samples to reduce bias and hallucination. For example, Learning Chain of Counterfactual Thought (Zhang et al., 2020) disentangles factual knowledge from reasoning via CoBRa and CoCT datasets; C-VQA (Zhang et al., 2024b) and CRIPP-VQA (Patel et al., 2022) construct benchmarks for counterfactual VQA in static and video settings, respectively; Counterfactual Vision and Language Learning (Abbasnejad et al., 2020) and Counterfactual Contrastive Learning (Zhang et al., 2024c) generate counterfactuals through structural causal models and perturbation strategies, while CounterCurate (Zhang et al., 2024a) improves compositional reasoning by augmenting training data with physically grounded examples and semantic counterfactuals using generative models. The second type focuses on inference with mechanisms such

as Counterfactual-based Saliency Maps (Wang et al., 2023) for contrastive visual explanation, DiG-IN (Augustin et al., 2024) for diffusion-guided latent edits, and Counterfactual VQA (Niu et al., 2021) for causal effect modeling. CFMM (Li et al., 2024) curates a manually labeled benchmark of presupposition-based counterfactual questions to systematically expose MLLMs’ limitations in hypothetical visual reasoning. Beyond counterfactual perturbations, MM-Verify (Sun et al., 2025) addresses a related failure mode in multimodal reasoning by introducing a verification model that checks chain-of-thought trajectories.

Structured Thinking with Images in Vision-Language Models. The concept of “thinking with images” was initially highlighted in OpenAI o3 (Achiam et al., 2023; OpenAI, 2025) and later explored in works like COGCOM (Qi et al., 2024) and GRIT (Fan et al., 2025). Recent datasets also highlight the importance of evaluating visual reasoning beyond raw perception. For example, VisCoT (Shao et al., 2024) provides visual-evidence for vqa, while datasets such as MSTI (Chen et al., 2024b) emphasize structured visual understanding across detection, entity grounding, and answer generation. Existing approaches can be broadly categorized into two classes. The first category includes GRIT, which combines natural language and bounding boxes via reinforcement learning; REFOCUS (Fu et al., 2025), which formulates visual editing as intermediate reasoning steps; COGCOM (Qi et al., 2024), which models reasoning as visual manipulations such as cropping and OCR; and VisionReasoner (Liu et al., 2025a), which unifies detection, segmentation, and counting under one framework. The second category emphasizes grounding quality. Fast-and-Slow Visual Agents (Sun et al., 2024) model dual-system reasoning. DeepEyes (Zheng et al., 2025) leverages reinforcement learning to train multimodal chains-of-thought and dynamically invoke zoom-in tools when visual evidence is ambiguous. MLLMs Know Where to Look (Zhang et al., 2025a) improves small-object perception by applying inference-time cropping strategies to highlight fine details. Chain-of-Focus (Zhang et al., 2025b) further adapts zoom-in operations through reinforcement learning, enabling multi-scale reasoning across cluttered scenes. Ground-R1 (Cao et al., 2025) enhances faithfulness by introducing explicit reward signals that align reasoning outputs with grounded evidence. V* (Wu & Xie, 2024) formulates guided visual search as a core cognitive mechanism to explore high-resolution images efficiently. Visual-RFT (Liu et al., 2025b) refines grounding via reinforcement fine-tuning. PAPO (Wang et al., 2025b) introduces an implicit perception loss into GRPO to jointly optimize perception and reasoning. TreeVGR (Wang et al., 2025a) explicitly supervises bounding-box generation with a dual IoU reward during RL, providing traceable evidence for visual grounded reasoning.

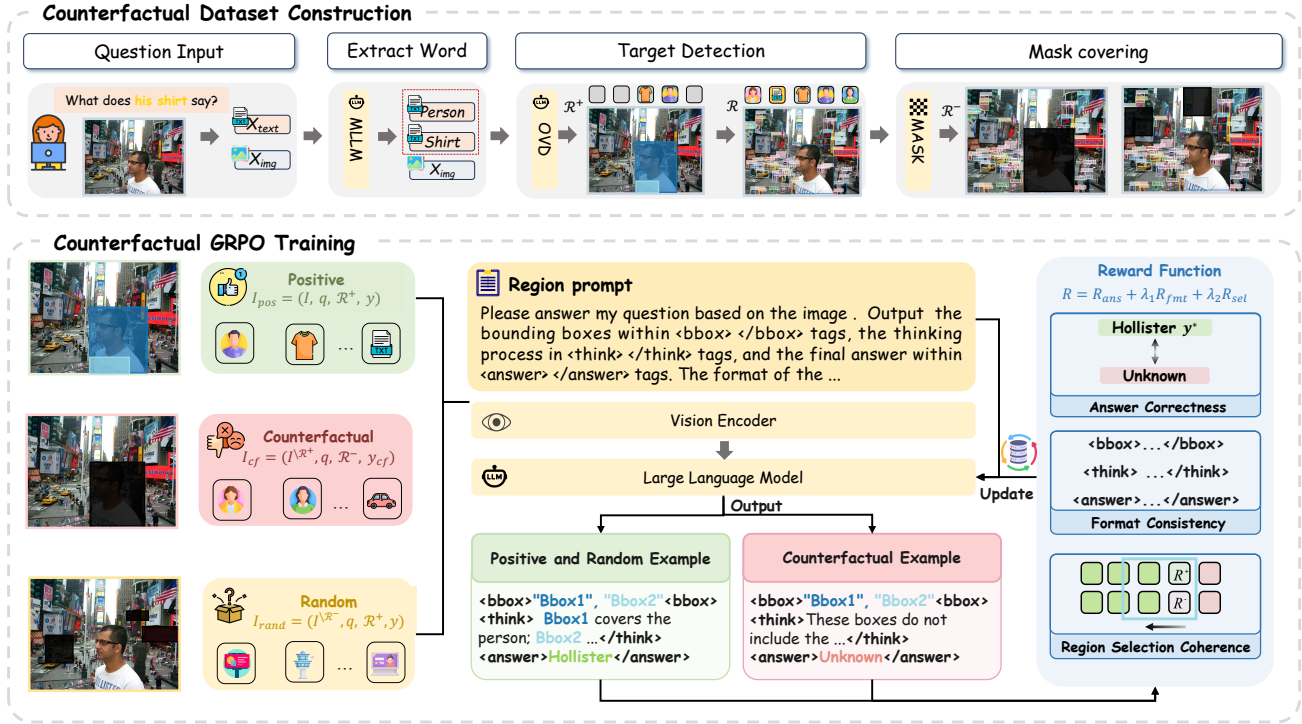


Figure 2. An overview of our counterfactual framework with three inputs: positive (full evidence), counterfactual (masked evidence), and random (masked irrelevant regions).

3. Method

In this section, we present the overall framework of DEFACTO, our counterfactual “thinking with images” approach. This section is organized into three parts: (1) the overall architecture and inference pipeline (Section 3.1); (2) the construction of counterfactual datasets via region masking and open-vocabulary filtering (Section 3.2); and (3) the reinforcement learning strategy with a tailored reward design that guides the model toward accurate answering, faithful reasoning, and their consistency (Section 3.3).

3.1. Overall Framework

DEFACTO is a vision-language reasoning framework that enforces region-level faithfulness in multimodal question answering. It is designed to teach models not only *where to look* in the image but also *when to abstain* if the necessary evidence is absent. By combining structured prompting with counterfactual supervision, DEFACTO aligns the reasoning process with visual evidence rather than spurious correlations.

As illustrated in Figure 2, given an image and a question explicitly specified in the prompt, the model is required to generate outputs in a unified structured format composed of three fields. The `<bbox>` field encodes one or multiple evidence regions as a JSON list of $[x1, y1, x2, y2]$ co-

ordinates, or outputs null when no valid visual grounding is available. The `<think>` field provides a concise reasoning process that must be explicitly grounded in the selected bounding boxes (if any) and directly related to the given question. The `<answer>` field delivers the final prediction. This structured format ensures that every reasoning trajectory is explicitly tied to visual evidence through bounding boxes and aligned with the model’s final answer. Training is based on three complementary supervision forms. In the positive case, evidence-bearing regions remain visible and the model is rewarded for selecting them and producing the correct answer. In the counterfactual case, these regions are masked, and the model is expected to abstain by outputting unknown. In the random-masking case, irrelevant regions are occluded to prevent shortcut learning from superficial mask patterns. Together, these three forms establish a consistent learning signal that requires both the reasoning path and the answer to be faithful to the underlying visual support.

3.2. Counterfactual Dataset Construction

Positive, Counterfactual, and Random Instances. Let $\mathcal{R} = \{r_1, r_2, \dots, r_n\}$ denote the set of candidate regions in an image I , obtained from a region proposal network (RPN) (Ren et al., 2015) together with OCR to cover both object-level and text-bearing regions. Among them, $\mathcal{R}^+ \subseteq \mathcal{R}$ represents the evidence regions that are critical to

answering the question q , while $\mathcal{R}^- = \mathcal{R} \setminus \mathcal{R}^+$ denotes the remaining irrelevant regions. Based on these definitions, we construct three complementary training instances:

$$\begin{aligned} I_{\text{pos}} &= (I, q, \mathcal{R}^+, y), \\ I_{\text{cf}} &= (I \setminus \mathcal{R}^+, q, \mathcal{R}^-, y_{\text{cf}}), \\ I_{\text{rand}} &= (I \setminus \mathcal{R}^-, q, \mathcal{R}^+, y). \end{aligned} \quad (1)$$

where y is the ground-truth answer, I_{pos} is the positive instance with evidence regions available, I_{cf} is the counterfactual instance where evidence regions are masked and the abstention label y_{cf} (e.g., “Unknown”) is required, and I_{rand} is the random-masking instance where irrelevant regions are occluded to prevent shortcut learning.

Construction Process. To automatically construct I_{pos} , I_{cf} , and I_{rand} without manual annotations, we follow three steps:

(1) Descriptor extraction. Given an image I and a question q , we employ a MLLM (Qwen2.5-VL (Bai et al., 2025)) to extract a set of key descriptors:

$$\text{MLLM}(I, q) = \mathcal{K}(q) = \{d_1, d_2, \dots, d_m\}, \quad (2)$$

where each d_i is a textual phrase (e.g., an object, attribute, or relation) that captures the visual concepts in I explicitly mentioned or implied by q . As illustrated in Fig. 2 (“Automate Evidence Masking”), for the question “What does his shirt say?”, the MLLM decomposes the query into descriptors such as “a man” and “man’s shirt” as the critical evidence.

(2) Evidence localization. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the set of candidate image regions. We employ the open-vocabulary detector DINO-X (Ren et al., 2024), which computes grounding scores $\text{OVD}(r, k)$ for each $r \in \mathcal{R}$ and $k \in \mathcal{K}(q)$. Based on these scores, the regions are partitioned into evidence and irrelevant sets:

$$\begin{aligned} \mathcal{R}^+ &= \left\{ r \in \mathcal{R} \mid \max_{k \in \mathcal{K}(q)} \text{OVD}(r, k) > \tau \right\}, \\ \mathcal{R}^- &= \mathcal{R} \setminus \mathcal{R}^+. \end{aligned} \quad (3)$$

where τ is a confidence threshold. In the street example, $\mathcal{R}^+$ corresponds to bounding boxes covering the signboard, while $\mathcal{R}^-$ contains all other regions.

(3) Instance generation. Once $\mathcal{R}^+$ and $\mathcal{R}^-$ are obtained, the positive, counterfactual, and random-masking instances are directly constructed as defined in Eq. 1.

For counterfactual dataset construction, we leverage a broad collection of visual question answering and document understanding benchmarks, including VQAv2 (Goyal

et al., 2017) for general visual question answering, ChartQA (Masry et al., 2022) for chart-based reasoning, and DocVQA (Mathew et al., 2021) for document understanding, among others. This diverse coverage ensures that counterfactual supervision is tested across natural images, scientific diagrams, documents, charts, tables, and multi-source reasoning tasks.

3.3. Counterfactual GRPO Training

Sequential Reasoning Formulation. We formulate the reasoning process of DEFACTO as a Markov Decision Process (MDP), where the model interacts with the question and image in a sequential manner. At each step, the state s_t encodes the multimodal context, including the input question, the image representation, and the history of previously predicted regions. The policy π_θ then outputs either a new bounding box that localizes question-relevant evidence or a special STOP token to terminate the process.

Formally, the state at step t is defined as

$$s_t = \{q, f_v(I), B_{<t}\}, \quad (4)$$

where q is the question, $f_v(I)$ the image representation, and $B_{<t}$ the set of bounding boxes predicted before step t . The rollout continues until STOP is emitted or the maximum step limit is reached, and the final answer is generated based on the accumulated trajectory.

Reward Design. To make training effective, we design three reward components. (1) *Answer Correctness Reward*: encourages correct answers in positive/random cases, rewards “Unknown” in counterfactual cases, and penalizes unsupported guesses. (2) *Format Consistency Reward*: ensures outputs strictly follow the required schema. (3) *Region Selection Coherence Reward*: promotes overlap with evidence regions $\mathcal{R}^+$ and penalizes overlap with irrelevant regions $\mathcal{R}^-$, with no reward in counterfactual cases.

The overall training signal combines these components into the composite reward in Eq. 5.

$$R = R_{\text{ans}} + \lambda_1 R_{\text{fmt}} + \lambda_2 R_{\text{sel}}, \quad (5)$$

1. Answer Correctness Reward. To enforce correct behavior across the three training forms, we define

$$R_{\text{ans}} = \begin{cases} \underbrace{\text{acc}(\hat{y}, y^*)}_{\text{reward "Unknown"}} - \underbrace{\gamma_{\text{unk}} \text{unk}(\hat{y})}_{\text{penalize "Unknown"}}, & t \in \{\text{pos}, \text{rand}\}, \\ \underbrace{\rho_{\text{unk}} \text{unk}(\hat{y})}_{\text{reward "Unknown"}} - \underbrace{\gamma_{\text{guess}} [1 - \text{unk}(\hat{y})]}_{\text{penalize guess}} - \underbrace{\gamma_{\text{corr}} \mathbf{1}[\hat{y} = y^*]}_{\text{penalize even if correct}}, & t = \text{cf}. \end{cases} \quad (6)$$

where $\text{acc}(\hat{y}, y^*) \in \{0, 1\}$ indicates answer correctness, and $\text{unk}(\hat{y}) \in \{0, 1\}$ indicates an “Unknown” response. Here $\gamma_{\text{unk}} > 0$ penalizes answering “Unknown” in positive or

random cases, $\rho_{\text{unk}} > 0$ rewards “Unknown” in counterfactual cases, $\gamma_{\text{guess}} > 0$ penalizes any concrete guess in counterfactual cases, and $\gamma_{\text{corr}} > \gamma_{\text{guess}}$ applies an even stronger penalty when the model outputs the correct answer y^* without access to evidence.

2. Format Consistency Reward. We encourage well-formed outputs with valid bounding box coordinates and the required response structure:

$$R_{\text{fmt}} = \begin{cases} \alpha, & \text{if output follows the required schema,} \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

Here, the “required schema” refers to the presence of `<think>...</think>` for the reasoning process, `<bbox>...</bbox>` for the predicted bounding boxes, and `<answer>...</answer>` for the final answer. In particular, the `<bbox>` field must contain well-formed bounding box coordinates in the format `[x1, y1, x2, y2]`, ensuring that the model explicitly grounds its predictions on localized visual regions.

3. Region Selection Coherence Reward. Let $B = \{\text{bbox}^1, \dots, \text{bbox}^k\}$ be the set of bounding boxes predicted before STOP. We define the overlap scores with evidence regions $\mathcal{R}^+$ and irrelevant regions $\mathcal{R}^-$ as

$$\phi^+(b) = \max_{r \in \mathcal{R}^+} \text{IoU}(b, r), \quad \phi^-(b) = \max_{r \in \mathcal{R}^-} \text{IoU}(b, r).$$

The reward is then defined as

$$R_{\text{sel}} = \begin{cases} \beta_{\text{pos}} \frac{1}{|B|} \sum_{b \in B} \phi^+(b) - \beta_{\text{neg}} \frac{1}{|B|} \sum_{b \in B} \phi^-(b), & t \in \{\text{pos}, \text{rand}\}, B \neq \emptyset, \\ -\gamma_{\emptyset}, & t \in \{\text{pos}, \text{rand}\}, B = \emptyset, \\ 0, & t = \text{cf}. \end{cases} \quad (8)$$

with $\beta_{\text{pos}}, \beta_{\text{neg}}, \gamma_{\emptyset} > 0$.

Training Strategy. We fine-tune Qwen2.5-VL with reinforcement learning, using Group Relative Policy Optimization (GRPO) and the composite reward in Eq. 6 and Eq. 8. GRPO compares multiple rollouts within a group and rewards each according to its improvement over the group average, eliminating the need for a value network and reducing variance. The objective is defined as:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q, \{o_i\}_{i=1}^G} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t}) - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} \parallel \pi_{\theta}] \right) \right], \quad (9)$$

where the importance ratio is

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad (10)$$

and the group-relative advantage is computed by normalizing rewards within each sampled group of G responses:

$$\hat{A}_{i,t} = \frac{R(o_i) - \text{mean}(\{R(o_1), \dots, R(o_G)\})}{\text{std}(\{R(o_1), \dots, R(o_G)\})}. \quad (11)$$

where we set the group size $G = 4$ to balance stability and exploration during training.

4. Experiment

4.1. Setup

Training Configuration. We train all models with the AdamW optimizer using a learning rate of 1×10^{-6} , $(\beta_1, \beta_2) = (0.9, 0.999)$, and $\epsilon = 1 \times 10^{-8}$. Training is performed with a global batch size of 8 and micro-batch size of 1 per GPU, combined with gradient accumulation steps of 2. Gradients are clipped to a maximum norm of 1.0 to ensure stability. We enable BF16 precision training. All experiments are conducted on 8 NVIDIA H100 GPUs with 80GB memory each, and models are trained for one epoch over the collected dataset.

Baselines. We compare DEFACTO against a broad set of recent approaches that explicitly incorporate visual reasoning into multimodal language models. Specifically, we include the QWEN2.5-VL (Bai et al., 2025), a strong pre-trained backbone widely used for visual understanding; VICROP (Zhang et al., 2025a), which improves small-object perception via inference-time cropping; GRIT (Fan et al., 2025), which integrates grounded reasoning traces through reinforcement learning; DEEPEYES (Zheng et al., 2025), which incentivizes models to call visual tools during reasoning; and VISUAL-SR1 (Li et al., 2025), which enhances step-by-step visual reasoning with self-refinement. This selection covers both state-of-the-art backbones and recent “thinking with images” algorithms for visual reasoning.

Benchmarks. Our evaluation spans a diverse collection of visual reasoning benchmarks. For general-purpose VQA, we use OKVQA (Marino et al., 2019), VQAv2 (Goyal et al., 2017), GQA (Hudson & Manning, 2019), VizWiz (Gurari et al., 2018), ScienceQA (Lu et al., 2022), and VSR (Liu et al., 2023). For document- and structure-centric evaluation, we adopt DocVQA (Mathew et al., 2021), ChartQA (Masry et al., 2022), InfoVQA (Mathew et al., 2022), DeepForm (Svetlichnaya, 2020), Kleister KLC (Stanisławek et al., 2021), and WikiTableQuestions (WTQ) (Pasupat & Liang, 2015). To test text-intensive reasoning, we include TextVQA (Singh et al., 2019), AI2D (Kembhavi et al., 2016), and STVQA (Biten et al., 2019). We further construct **DeFacto-1.5K** by sampling 100 examples from each of the above evaluation benchmarks and manually annotating question-relevant evidence regions. For each question, annotators label one or multiple key bounding boxes that support the correct answer. In addition, we evaluate on more benchmarks, including OCRBench (Liu et al., 2024c), MMStar (Chen et al., 2024a), MMMU (Yue et al., 2024), MMB_{1.1} (Liu et al., 2024b), and POPE (Li

Table 1. Results on General VQA Benchmarks (accuracy, %). Δ indicates improvements of DeFacto over Qwen2.5-VL 7B.

Model	Backbone	VQAv2	OKVQA	GQA	SciQA	VizWiz	VSR
Qwen2.5-VL	Qwen2.5-VL-7B	<u>67.0</u>	<u>58.9</u>	<u>53.2</u>	83.0	54.1	63.6
ViCrop	LLaVA-1.5 (Vicuna-7B)	–	–	–	82.1	64.7	–
GRIT	Qwen2.5-VL-3B	64.7	43.1	54.8	59.8	39.4	62.4
DeepEyes	Qwen2.5-VL-7B	–	40.0	45.5	49.7	32.2	43.5
Visual-SR1	Qwen2.5-VL-7B	64.7	49.0	52.3	88.0	35.8	<u>66.4</u>
DeFacto (Ours)	Qwen2.5-VL-7B	72.1	61.7	63.9	<u>83.6</u>	<u>61.4</u>	71.0
Δ (vs Qwen2.5-VL 7B)	–	+5.1	+2.8	+10.7	+0.6	+7.3	+7.4

Table 2. Results on Document VQA and Scene Text-centric Benchmarks (accuracy, %). Δ indicates improvements of DeFacto over Qwen2.5-VL 7B.

Model	Document VQA						Scene Text-centric		
	DocVQA	ChartQA	InfoVQA	DeepForm	KLC	WTQ	STVQA	TextVQA	AI2D
Qwen2.5-VL	<u>92.0</u>	<u>74.4</u>	71.5	27.9	36.5	62.3	67.9	<u>79.1</u>	69.5
GRIT	<u>57.0</u>	<u>24.6</u>	43.7	17.9	17.8	32.0	64.7	<u>60.8</u>	<u>75.5</u>
ViCrop	31.8	50.8	52.4	20.3	29.1	51.5	<u>72.4</u>	60.8	68.8
DeepEyes	67.1	–	39.9	–	32.2	51.3	45.9	40.4	37.0
Visual-SR1	84.9	71.7	<u>76.5</u>	50.8	<u>36.6</u>	71.3	75.9	70.1	69.0
DeFacto (Ours)	94.0	82.1	79.1	<u>45.6</u>	38.9	<u>63.7</u>	71.2	82.9	76.1
Δ (vs Qwen2.5-VL 7B)	+2.0	+7.7	+7.6	+17.7	+2.4	+1.4	+3.3	+3.8	+6.6

et al., 2023b), where DEFACTo achieves competitive performance against both leading closed-source models (e.g., GPT-4o (Microsoft, 2024), Gemini-1.5-Pro (Team et al., 2024)) and recent thinking-with-images approaches, as summarized in Table 3.

Table 3. Comparison with representative closed-source and thinking-with-images models.

Model	OCRB	MMStar	MMMU	MMB _{1.1}	POPE
GPT-4o	736	63.9	69.2	82.2	–
Gemini-1.5-Pro	<u>754</u>	–	<u>62.2</u>	–	–
DeepEyes	636	43.6	44.1	29.4	87.7
Chain-of-Focus	632	58.1	46.1	75.3	88.4
Pixel Reasoner	597	62.9	52.5	78.5	87.8
Visual-SR1	449	62.8	57.2	77.4	86.0
DeFacto (Ours)	871	<u>63.2</u>	56.6	<u>81.2</u>	88.6

4.2. Main results

Results on General VQA Benchmarks. Table 1 reports results on six widely used general-domain VQA benchmarks. Built upon the Qwen2.5-VL-7B backbone, DEFACTo achieves the best performance on four out of six benchmarks, demonstrating consistent gains in both general reasoning and fine-grained visual understanding. Compared with the Qwen2.5-VL baseline, DEFACTo improves accuracy from 67.0 to 72.1 on VQAv2 (+5.1), from 58.9 to 61.7 on OKVQA (+2.8), and from 53.2 to 63.9 on GQA (+10.7). Notably, DEFACTo also delivers substantial improvements on perception-intensive datasets, reaching 83.6

on SciQA (+0.6), 61.4 on VizWiz (+7.3), and 71.0 on VSR (+7.4). Overall, DEFACTo consistently surpasses representative thinking-with-images baselines (e.g., GRIT, DeepEyes, Visual-SR1, and ViCrop where applicable), establishing strong performance across all reported general VQA benchmarks.

Performance on Document and Text-centric Benchmarks. Table 2 reports results on document VQA and scene text-centric benchmarks. DEFACTo achieves strong overall performance, improving over Qwen2.5-VL on every benchmark, including DocVQA (+2.0), ChartQA (+7.7), InfoVQA (+7.6), DeepForm (+17.7), KLC (+2.4), WTQ (+1.4), STVQA (+3.3), TextVQA (+3.8), and AI2D (+6.6). DEFACTo attains the best results on six out of nine benchmarks (DocVQA, ChartQA, InfoVQA, KLC, TextVQA, and AI2D), and remains competitive on the remaining three (DeepForm, WTQ, and STVQA). These results suggest that DEFACTo effectively enhances evidence-grounded understanding for document and text-centric visual reasoning.

Faithful reasoning evaluation. Fig. 3 visualizes the evidence-answer consistency results on **DeFacto-1.5K** (100 samples per benchmark) using metric-wise heatmaps over datasets and methods. We report grounding quality and task performance measured by mAP/AP50/AP75, IoU, and answer accuracy. Following the standard detection interpretation, we adopt mAP as the core faithfulness metric, which jointly reflects evidence localization and answer correctness: localization is quantified by IoU between the predicted ev-

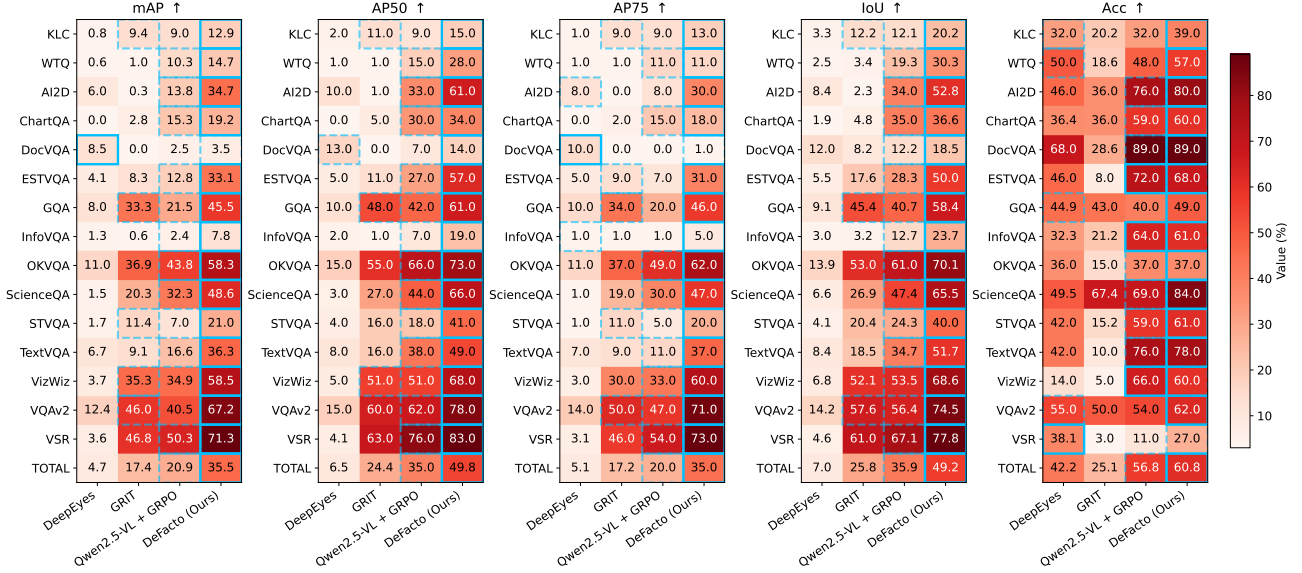


Figure 3. Heatmap visualization of faithful reasoning metrics across DeFacto-1.5K.

Table 4. Ablation study with component breakdown on representative benchmarks (accuracy, %).

Setting	SFT	CF Align.	RandMask	GRPO	CF Reward	VQAv2	OKVQA	SciQA	VSR	DVQA	TVQA
(1) SFT (no CF)	✓	-	-	-	-	61.2	42.0	82.7	54.5	51.9	56.0
(2) SFT + CF alignment	✓	✓	✓	-	-	66.5	55.7	84.7	53.7	84.3	73.0
(3) GRPO (no CF reward)	✓	✓	✓	✓	-	70.4	56.9	85.9	58.4	85.4	72.8
(4) DeFacto (full)	✓	✓	✓	✓	✓	72.1	61.7	83.6	71.0	94.0	82.9
Δ (4) - (3)	-	-	-	-	-	+1.7	+4.8	-2.3	+12.6	+8.6	+10.1

idence box and the ground-truth region, while the classification signal is defined by whether the predicted answer is correct. For each dataset and metric, blue solid boxes indicate the best-performing method, while blue dashed boxes denote the second best. Overall, DEFAC TO consistently yields the strongest evidence-grounded reasoning across benchmarks, achieving the best **TOTAL** scores (mAP 35.5, AP50 49.8, AP75 35.0, IoU 49.2) together with the highest accuracy (60.8). To dissect where these gains come from, we conduct a fine-grained failure-mode analysis on DeFacto-1.5K (Table 5). DEFAC TO substantially reduces *Mislocalized & Wrong* and *Spurious Correct* compared with GRIT (43.5/11.6 vs. 21.5/5.6), and achieves the lowest *Faithful but Wrong* rate (1.3). We further test counterfactual robustness by masking the human-annotated evidence or replacing the image entirely, and DEFAC TO achieves significantly higher abstention accuracy in both cases (64.1 and 61.4), confirming that its correctness genuinely depends on the visual evidence rather than language priors.

4.3. Ablation Study

We compare four training settings on Qwen2.5-VL 7B: (i) **SFT (no CF)**: trained only on original data. (ii) **SFT (CF**

Table 5. Region-level faithfulness analysis on DeFacto-1.5K.

Metric	DeepEyes	GRIT	Ours
Mislocalized & Wrong ↓	23.5	43.5	21.5
Spurious Correct ↓	11.0	11.6	5.6
Faithful but Wrong ↓	1.8	30.1	1.3
Masked Evidence Abstention ↑	33.1	46.1	64.1
Image Replacement Abstention ↑	40.3	46.3	61.4

alignment): trained on original + counterfactual data, but counterfactuals are supervised only with the “Unknown” label, together with random-masking. (iii) **GRPO (no CF reward)**: “No CF reward” means that GRPO training uses only the first term of Eq. 6 (answer correctness) and the format reward. (iv) **DeFacto (full)**: our complete framework with all three rewards. In addition, we conduct a controlled ablation study on our **DeFacto-1.5K** benchmark to evaluate evidence-answer consistency under these training variants. Specifically, we report detection-style faithfulness metrics (mAP/AP50/AP75) computed against human-annotated evidence boxes, together with answer accuracy.

Effect of Counterfactual Supervision. The first two rows in Table 4 show that introducing counterfactual data with abstention alignment brings clear gains over standard SFT across most benchmarks. In particular, accuracy improves by +5.3% on VQAv2 (61.2→66.5) and +13.7% on OKVQA (42.0→55.7), with additional improvements on SciQA (+2.0%) and substantial gains on text-centric tasks such as DVQA (DocVQA) (+32.4%) and TVQA (TextVQA) (+17.0%). These results suggest that counterfactual supervision effectively strengthens evidence–answer consistency and reduces spurious shortcut behaviors.

Effect of Reinforcement Learning. Compared with the GRPO baseline without counterfactual reward, DeFACTO (CF reward + GRPO) achieves the strongest overall performance. As shown in Table 4, incorporating counterfactual rewards yields consistent improvements on five out of six benchmarks, including +1.7% on VQAv2, +4.8% on OKVQA, and a remarkable +12.6% on VSR. Particularly large gains are observed on document and text-centric evaluations, with +8.6% on DocVQA and +10.1% on TextVQA.

Table 6. Performance of Different Training Variants on the DeFacto-1.5K Validation Set.

Model Variant	mAP	AP50	AP75	Accuracy (%)
Qwen2.5-VL (Base)	0.7	1.4	0.7	52.5
SFT (no CF)	0.0	0.0	0.0	53.8
SFT (CF alignment)	2.9	4.4	1.7	57.0
GRPO (no CF reward)	20.9	35.0	20.0	56.8
DeFacto (full)	35.5	49.8	35.0	60.8

Faithfulness Evaluation. Table 6 compares the Qwen2.5-VL base model and several training variants on **DeFacto-1.5K**, where we evaluate evidence–answer consistency using detection-style faithfulness metrics (mAP/AP50/AP75) together with answer accuracy. In particular, our full framework reaches the highest mAP (35.5), AP50 (49.8), and AP75 (35.0), while also obtaining the highest accuracy (60.8). Compared to the base model and different component combinations, these results indicate that DeFACTO most effectively improves evidence–answer consistency by jointly optimizing evidence selection and final prediction.

5. Conclusion

In this work, we introduced *DeFacto*, a vision-language reasoning framework grounded in counterfactual supervision and designed to enforce region-faithful reasoning and abstention behavior when critical evidence is missing. To enable this paradigm, we proposed an automatic pipeline that leverages language model parsing, open-vocabulary detection, and OCR to mask question-relevant regions without manual annotation, and used it to construct a counterfactual dataset of about 100k images, together with a human-

verified benchmark for evaluating evidence-grounded reasoning. Extensive experiments across diverse benchmarks demonstrate that *DeFacto* consistently improves both answer accuracy and visual grounding faithfulness over strong baselines, while our ablation studies confirm the necessity of counterfactual training and region-level reward design. We note several limitations: our region proposal relies on publicly available detectors (e.g., RPN, OCR, and open-vocabulary models), which may introduce occasional errors but can be alleviated as stronger detectors become available; our counterfactual dataset, while sufficient for controlled experiments, remains modest compared to large-scale pre-training corpora; and our evaluation has so far focused on static images, leaving the extension to videos and temporal reasoning as an open direction. We believe these findings open new directions for integrating counterfactual supervision into multimodal reasoning systems, with potential extensions to video understanding and embodied AI.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Abbasnejad, E., Teney, D., Parvaneh, A., Shi, J., and Hengel, A. v. d. Counterfactual vision and language learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10044–10054, 2020.
- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.
- Augustin, M., Neuhaus, Y., and Hein, M. Dig-in: Diffusion guidance for investigating networks-uncovering classifier differences neuron visualisations and visual counterfactual explanations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11093–11103, 2024.
- Bai, J., Bai, S., Chu, Y., Cui, Z., Dang, K., Deng, X., Fan, Y., Ge, W., Han, Y., Huang, F., et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023.
- Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang,

- K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., Yang, M., Li, Z., Wan, J., Wang, P., Ding, W., Fu, Z., Xu, Y., Ye, J., Zhang, X., Xie, T., Cheng, Z., Zhang, H., Yang, Z., Xu, H., and Lin, J. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Biten, A. F., Tito, R., Mafla, A., Gomez, L., Rusinol, M., Valveny, E., Jawahar, C., and Karatzas, D. Scene text visual question answering. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 4291–4301, 2019.
- Cao, M., Zhao, H., Zhang, C., Chang, X., Reid, I., and Liang, X. Ground-r1: Incentivizing grounded visual reasoning via reinforcement learning. *arXiv preprint arXiv:2505.20272*, 2025.
- Chen, L., Li, J., Dong, X., Zhang, P., Zang, Y., Chen, Z., Duan, H., Wang, J., Qiao, Y., Lin, D., et al. Are we on the right way for evaluating large vision-language models? *arXiv preprint arXiv:2403.20330*, 2024a.
- Chen, Z., Lin, H., Luo, Z., Cheng, M., Ma, J., and Chen, G. Cofipara: A coarse-to-fine paradigm for multimodal sarcasm target identification with large multimodal models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9663–9687, 2024b.
- Chen, Z., Wang, W., Tian, H., Ye, S., Gao, Z., Cui, E., Tong, W., Hu, K., Luo, J., Ma, Z., et al. How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites. *Science China Information Sciences*, 67(12):220101, 2024c.
- Dettmers, T., Pagnoni, A., Holtzman, A., and Zettlemoyer, L. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36:10088–10115, 2023.
- Fan, Y., He, X., Yang, D., Zheng, K., Kuo, C.-C., Zheng, Y., Narayanaraju, S. J., Guan, X., and Wang, X. E. Grit: Teaching mllms to think with images. *arXiv preprint arXiv:2505.15879*, 2025.
- Fu, X., Liu, M., Yang, Z., Corring, J., Lu, Y., Yang, J., Roth, D., Florencio, D., and Zhang, C. Refocus: Visual editing as a chain of thought for structured image understanding. *arXiv preprint arXiv:2501.05452*, 2025.
- Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., and Parikh, D. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6904–6913, 2017.
- Gurari, D., Li, Q., Stangl, A. J., Guo, A., Lin, C., Grauman, K., Luo, J., and Bigham, J. P. Vizwiz grand challenge: Answering visual questions from blind people. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3608–3617, 2018.
- Hudson, D. A. and Manning, C. D. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6700–6709, 2019.
- Islam, N., Islam, Z., and Noor, N. A survey on optical character recognition system. *arXiv preprint arXiv:1710.05703*, 2017.
- Kembhavi, A., Salvato, M., Kolve, E., Seo, M., Hajishirzi, H., and Farhadi, A. A diagram is worth a dozen images. In *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part IV 14*, pp. 235–251. Springer, 2016.
- Li, J., Li, D., Savarese, S., and Hoi, S. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023a.
- Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W. X., and Wen, J.-R. Evaluating object hallucination in large vision-language models. *arXiv preprint arXiv:2305.10355*, 2023b.
- Li, Y., Tian, W., Jiao, Y., Chen, J., and Jiang, Y.-G. Eyes can deceive: Benchmarking counterfactual reasoning abilities of multi-modal large language models. *arXiv preprint arXiv:2404.12966*, 3, 2024.
- Li, Z., Yu, W., Huang, C., Liu, R., Liang, Z., Liu, F., Che, J., Yu, D., Boyd-Graber, J., Mi, H., et al. Self-rewarding vision-language model via reasoning decomposition. *arXiv preprint arXiv:2508.19652*, 2025.
- Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- Liu, H., Li, C., Li, Y., and Lee, Y. J. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26296–26306, 2024a.
- Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z., et al. Mmbench: Is your multi-modal model an all-around player? In *European conference on computer vision*, pp. 216–233. Springer, 2024b.

- Liu, Y., Li, Z., Huang, M., Yang, B., Yu, W., Li, C., Yin, X.-C., Liu, C.-L., Jin, L., and Bai, X. Ocrbench: on the hidden mystery of ocr in large multimodal models. *Science China Information Sciences*, 67(12):220102, 2024c.
- Liu, Y., Qu, T., Zhong, Z., Peng, B., Liu, S., Yu, B., and Jia, J. Visionreasoner: Unified visual perception and reasoning via reinforcement learning. *arXiv preprint arXiv:2505.12081*, 2025a.
- Liu, Z., Sun, Z., Zang, Y., Dong, X., Cao, Y., Duan, H., Lin, D., and Wang, J. Visual-rft: Visual reinforcement fine-tuning. *arXiv preprint arXiv:2503.01785*, 2025b.
- Lu, P., Mishra, S., Xia, T., Qiu, L., Chang, K.-W., Zhu, S.-C., Tafjord, O., Clark, P., and Kalyan, A. Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in Neural Information Processing Systems*, 35:2507–2521, 2022.
- Marino, K., Rastegari, M., Farhadi, A., and Mottaghi, R. Ok-vqa: A visual question answering benchmark requiring external knowledge. In *Proceedings of the IEEE/cvf conference on computer vision and pattern recognition*, pp. 3195–3204, 2019.
- Masry, A., Long, D. X., Tan, J. Q., Joty, S., and Hoque, E. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. *arXiv preprint arXiv:2203.10244*, 2022.
- Mathew, M., Karatzas, D., and Jawahar, C. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 2200–2209, 2021.
- Mathew, M., Bagal, V., Tito, R., Karatzas, D., Valveny, E., and Jawahar, C. Infographicvqa. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 1697–1706, 2022.
- Microsoft. Introducing gpt-4o-2024-08-06 api with structured outputs on azure. <https://techcommunity.microsoft.com/blog/azure-ai-services-blog/introducing-gpt-4o-2024-08-06-api-with-structured-outputs-on-azure/4232684>, 2024. Accessed: 2025-03-07.
- Niu, Y., Tang, K., Zhang, H., Lu, Z., Hua, X.-S., and Wen, J.-R. Counterfactual vqa: A cause-effect look at language bias. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 12700–12710, 2021.
- OpenAI. Thinking with images, 2025. URL <https://openai.com/index/thinking-with-images/>. Accessed: 2025-08-06.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Pasupat, P. and Liang, P. Compositional semantic parsing on semi-structured tables. *arXiv preprint arXiv:1508.00305*, 2015.
- Patel, M., Gokhale, T., Baral, C., and Yang, Y. Crippvqa: Counterfactual reasoning about implicit physical properties via video question answering. *arXiv preprint arXiv:2211.03779*, 2022.
- Peng, Z., Wang, W., Dong, L., Hao, Y., Huang, S., Ma, S., and Wei, F. Kosmos-2: Grounding multimodal large language models to the world. *arXiv preprint arXiv:2306.14824*, 2023.
- Qi, J., Ding, M., Wang, W., Bai, Y., Lv, Q., Hong, W., Xu, B., Hou, L., Li, J., Dong, Y., et al. Cogcom: A visual language model with chain-of-manipulations reasoning. *arXiv preprint arXiv:2402.04236*, 2024.
- Ren, S., He, K., Girshick, R., and Sun, J. Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems*, 28, 2015.
- Ren, T., Chen, Y., Jiang, Q., Zeng, Z., Xiong, Y., Liu, W., Ma, Z., Shen, J., Gao, Y., Jiang, X., Chen, X., Song, Z., Zhang, Y., Huang, H., Gao, H., Liu, S., Zhang, H., Li, F., Yu, K., and Zhang, L. Dino-x: A unified vision model for open-world object detection and understanding, 2024. URL <https://arxiv.org/abs/2411.14347>.
- Shao, H., Qian, S., Xiao, H., Song, G., Zong, Z., Wang, L., Liu, Y., and Li, H. Visual cot: Advancing multimodal language models with a comprehensive dataset and benchmark for chain-of-thought reasoning. *Advances in Neural Information Processing Systems*, 37:8612–8642, 2024.
- Singh, A., Natarajan, V., Shah, M., Jiang, Y., Chen, X., Batra, D., Parikh, D., and Rohrbach, M. Towards vqa models that can read. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8317–8326, 2019.
- Stanisławek, T., Galiński, F., Wróblewska, A., Lipiński, D., Kaliska, A., Rosalska, P., Topolski, B., and Biecek, P. Kleister: key information extraction datasets involving long documents with complex layouts. In *International Conference on Document Analysis and Recognition*, pp. 564–579. Springer, 2021.

- Su, A., Wang, H., Ren, W., Lin, F., and Chen, W. Pixel reasoner: Incentivizing pixel-space reasoning with curiosity-driven reinforcement learning. *arXiv preprint arXiv:2505.15966*, 2025.
- Sun, G., Jin, M., Wang, Z., Wang, C.-L., Ma, S., Wang, Q., Geng, T., Wu, Y. N., Zhang, Y., and Liu, D. Visual agents as fast and slow thinkers. *arXiv preprint arXiv:2408.08862*, 2024.
- Sun, L., Liang, H., Wei, J., Yu, B., Li, T., Yang, F., Zhou, Z., and Zhang, W. Mm-verify: Enhancing multimodal reasoning with chain-of-thought verification, 2025. URL <https://arxiv.org/abs/2502.13383>.
- Svetlichnaya, S. Deepform: Understand structured documents at scale. 2020.
- Team, G., Anil, R., Borgeaud, S., Alayrac, J.-B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A. M., Hauth, A., Millican, K., et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- Team, G., Georgiev, P., Lei, V. I., Burnell, R., Bai, L., Gulati, A., Tanzer, G., Vincent, D., Pan, Z., Wang, S., et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- Wang, H., Li, X., Huang, Z., Wang, A., Wang, J., Zhang, T., Zheng, J., Bai, S., Kang, Z., Feng, J., et al. Traceable evidence enhanced visual grounded reasoning: Evaluation and methodology. *arXiv preprint arXiv:2507.07999*, 2025a.
- Wang, X., Wang, Z., Weng, H., Guo, H., Zhang, Z., Jin, L., Wei, T., and Ren, K. Counterfactual-based saliency map: Towards visual contrastive explanations for neural networks. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 2042–2051, 2023.
- Wang, Z., Guo, X., Stoica, S., Xu, H., Wang, H., Ha, H., Chen, X., Chen, Y., Yan, M., Huang, F., et al. Perception-aware policy optimization for multimodal reasoning. *arXiv preprint arXiv:2507.06448*, 2025b.
- Wu, P. and Xie, S. V?: Guided visual search as a core mechanism in multimodal llms. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13084–13094, 2024.
- Wu, Z., Chen, X., Pan, Z., Liu, X., Liu, W., Dai, D., Gao, H., Ma, Y., Wu, C., Wang, B., et al. Deepseek-vl2: Mixture-of-experts vision-language models for advanced multimodal understanding. *arXiv preprint arXiv:2412.10302*, 2024.
- Xu, T., Chen, G., Li, Y., Xi, Y., Mu, Z., Wang, R., Zhang, T., Gao, H., and Chen, F. Ouro: A self-bootstrapped framework for enhancing multimodal scene understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 18240–18251, 2025a.
- Xu, T., Xu, S., Chen, X., Chen, F., and Li, H. Multi-core token mixer: a novel approach for underwater image enhancement. *Machine Vision and Applications*, 36(2): 1–16, 2025b.
- Yan, S., Tong, J., Xue, H., Tang, X., Wang, Y., Shi, K., Zhang, G., Li, R., and Zou, Y. Act wisely: Cultivating meta-cognitive tool use in agentic multimodal models. *arXiv preprint arXiv:2604.08545*, 2026.
- Yue, X., Ni, Y., Zhang, K., Zheng, T., Liu, R., Zhang, G., Stevens, S., Jiang, D., Ren, W., Sun, Y., et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9556–9567, 2024.
- Zhang, J., Cai, M., Xie, T., and Lee, Y. J. Countercurate: Enhancing physical and semantic visio-linguistic compositional reasoning via counterfactual examples. *arXiv preprint arXiv:2402.13254*, 2024a.
- Zhang, J., Khayatkhoei, M., Chhikara, P., and Ilievski, F. Mllms know where to look: Training-free perception of small visual details with multimodal llms. *arXiv preprint arXiv:2502.17422*, 2025a.
- Zhang, L., Zhai, X., Zhao, Z., Zong, Y., Wen, X., and Zhao, B. What if the tv was off? examining counterfactual reasoning abilities of multi-modal language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 21853–21862, 2024b.
- Zhang, X., Gao, Z., Zhang, B., Li, P., Zhang, X., Liu, Y., Yuan, T., Wu, Y., Jia, Y., Zhu, S.-C., et al. Chain-of-focus: Adaptive visual search and zooming for multimodal reasoning via rl. *arXiv preprint arXiv:2505.15436*, 2025b.
- Zhang, Y., Jiang, M., and Zhao, Q. Learning chain of counterfactual thought for bias-robust vision-language reasoning. In *European Conference on Computer Vision*, pp. 334–351. Springer, 2024c.
- Zhang, Z., Zhao, Z., Lin, Z., He, X., et al. Counterfactual contrastive learning for weakly-supervised vision-language grounding. *Advances in Neural Information Processing Systems*, 33:18123–18134, 2020.

Zheng, Z., Yang, M., Hong, J., Zhao, C., Xu, G., Yang, L., Shen, C., and Yu, X. Deepeyes: Incentivizing” thinking with images” via reinforcement learning. *arXiv preprint arXiv:2505.14362*, 2025.

Zhu, D., Chen, J., Shen, X., Li, X., and Elhoseiny, M. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*, 2023.